

paper001

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Abstract To be drafted.

Introduction

1. Introduction to gamma implications in the brain.
2. Summary of SOTA of modelling gamma.
3. We train Gamma for the first time.
4. It brings sparsity benefits by architecture.

Methods

Summary of single neuron equations

The model specialises in to two populations — E driven by excitation and inhibition, I by excitation only

$$C_m^E \frac{dV^E}{dt} = -g_L^E(V^E - E_L) - g_e^E(V^E - E_e) - g_i^E(V^E - E_i) \quad (1)$$

$$C_m^I \frac{dV^I}{dt} = -g_L^I(V^I - E_L) - g_e^I(V^I - E_e) \quad (2)$$

1. Summary of network architecture.
2. Defense of fixed W_{ei} W_{ie} weights.
3. Summary of rhythmicity metric with literature review.
4. Summary of training.
5. Summary of gradient damping method for the PING case.

Results

Paragraphs on recurrent loop creates Gamma. References Figure 1.

Paragraphs on supercritical Hopf bifurcation. References Figure 2.

Paragraphs on PING uses fewer E spikes. References Figure 3.

Paragraphs on sparsity is architecture enforced, not learned. References Figure 4 and 5.

Paragraphs on one spike per cell per cycle. References Figure 6, 7 and 8.

Paragraphs on the gating is dynamical and robust. References Figure 9, 10 and 11.

Paragraphs on real-time streaming classification. References Figure 12 and 13.

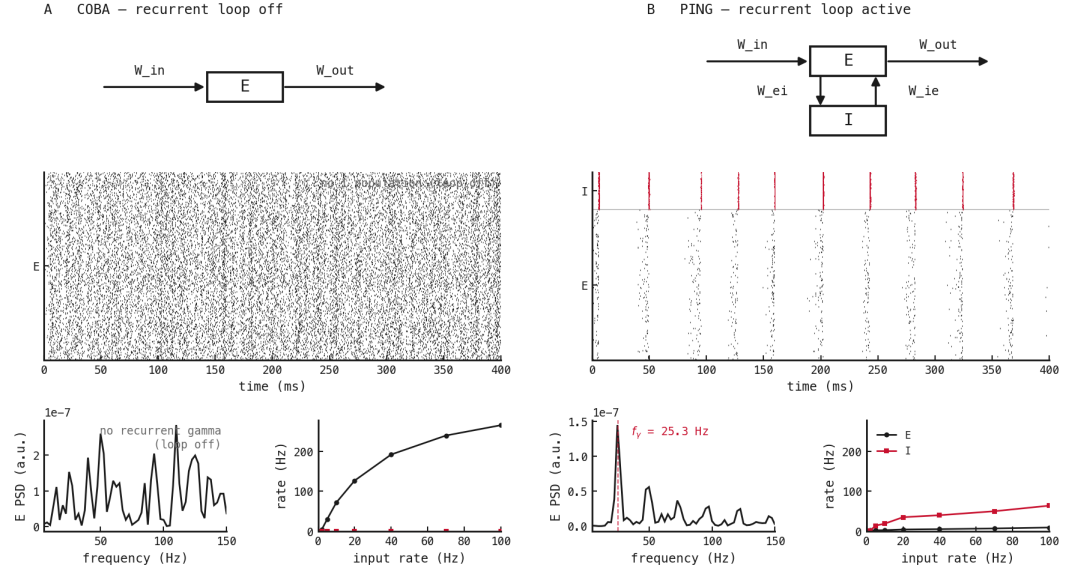
Discussion

To be drafted.

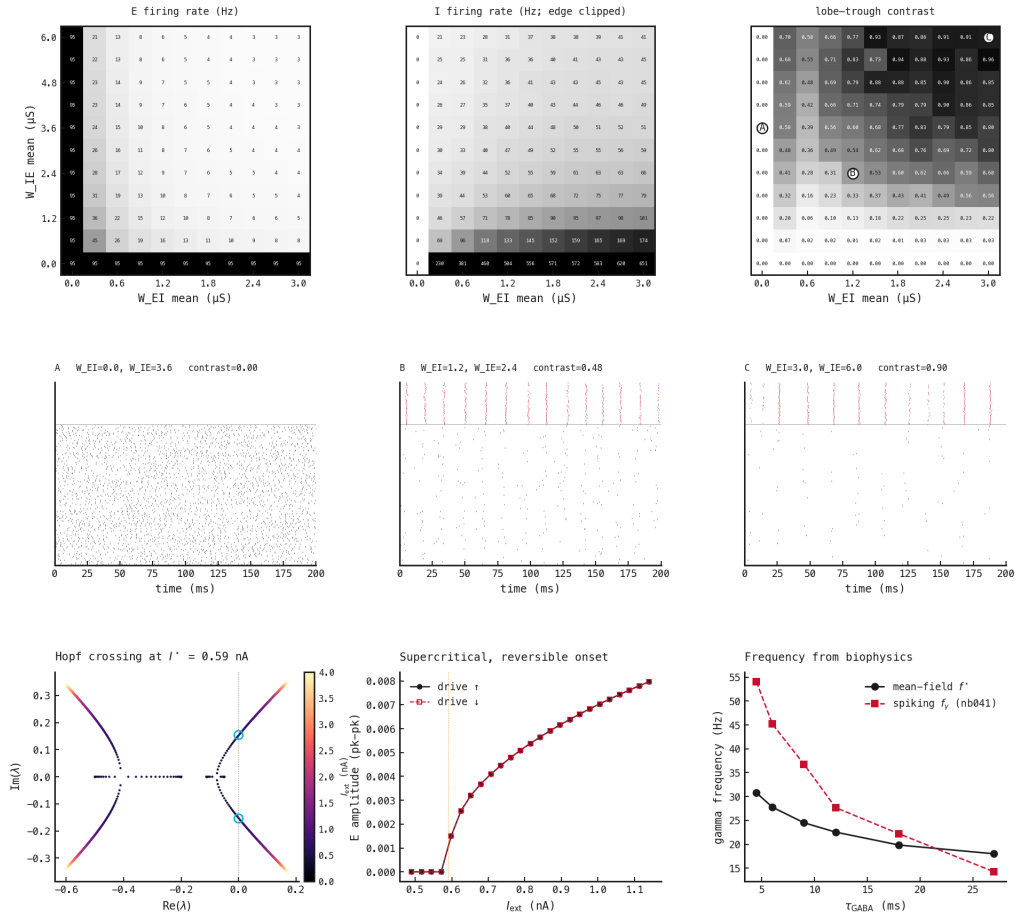
Conclusion

To be drafted.

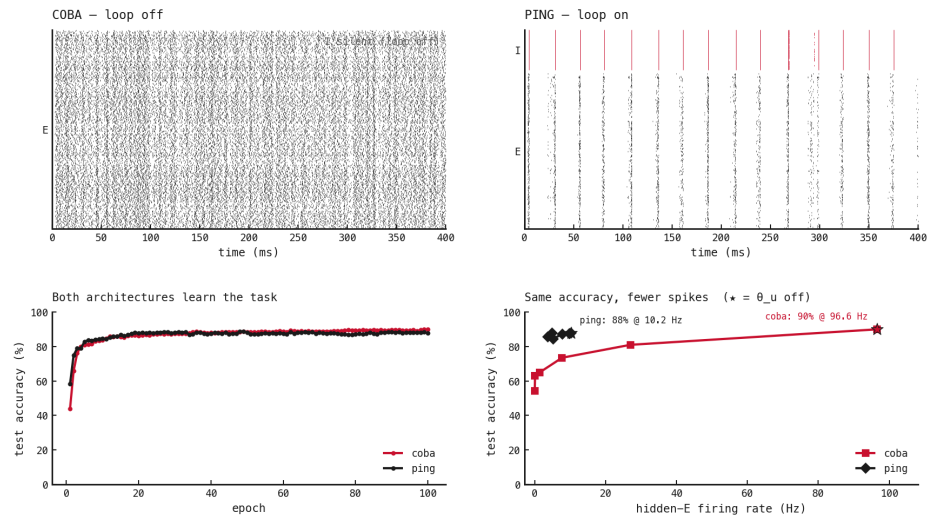
Figures



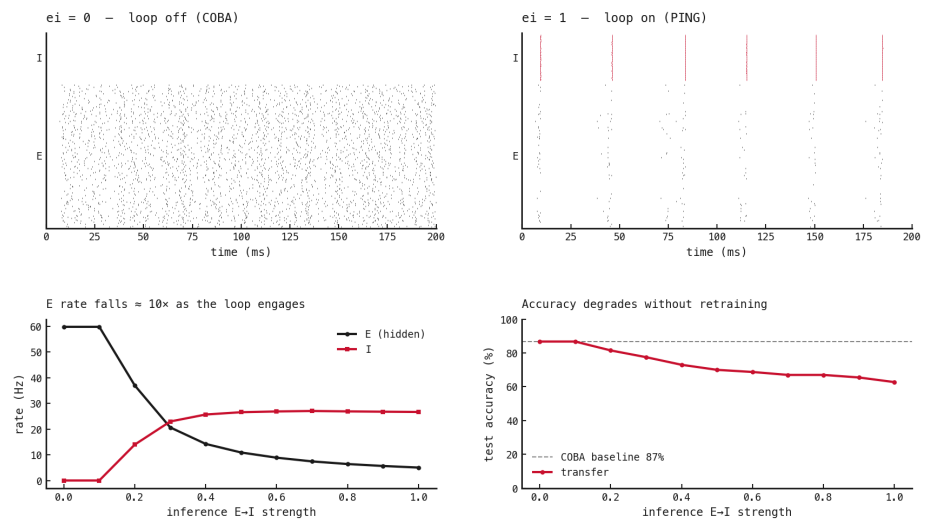
Appendix 0 Figure 1



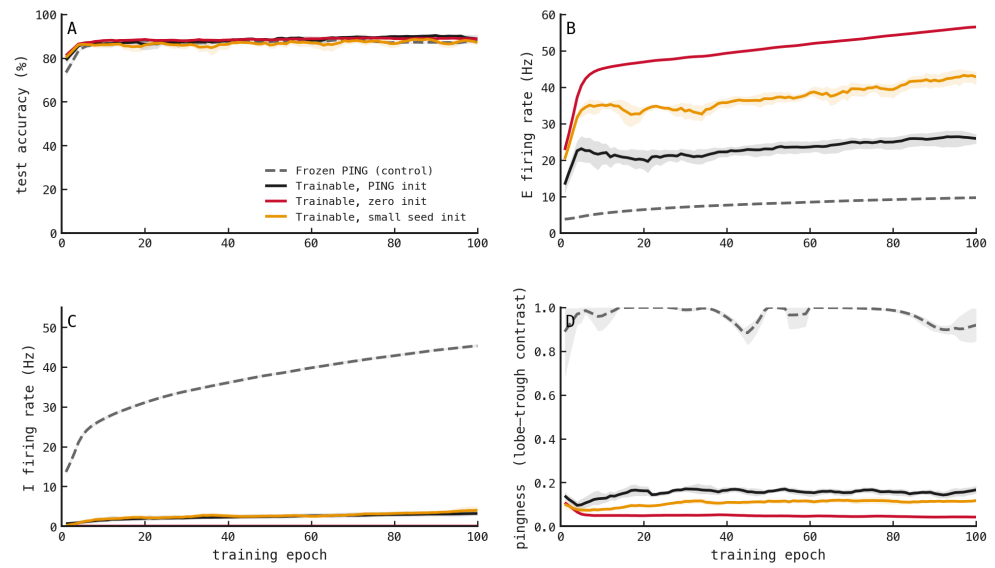
Appendix 0 Figure 2



Appendix 0 Figure 3



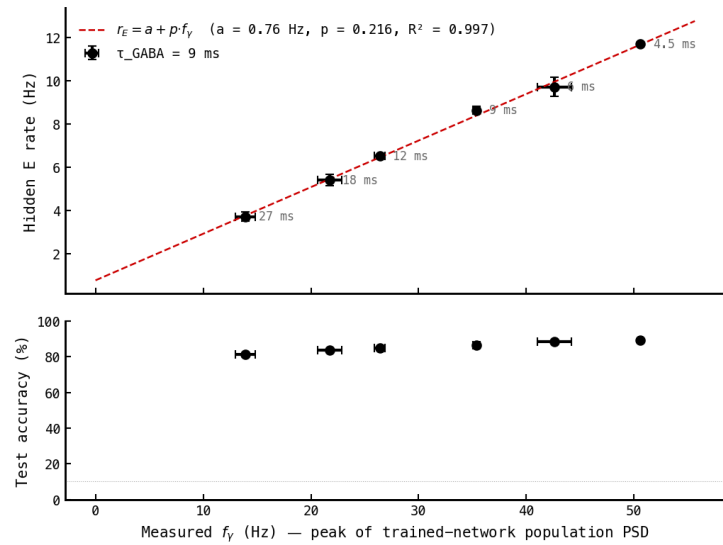
Appendix 0 Figure 4



nb049-curves

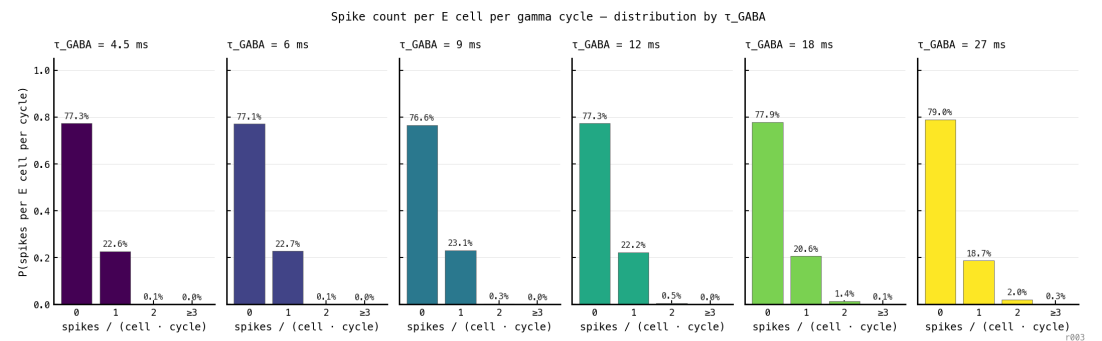
Appendix 0 Figure 5

Retrained τ_{GABA} sweep — post-training E rate tracks measured f_y

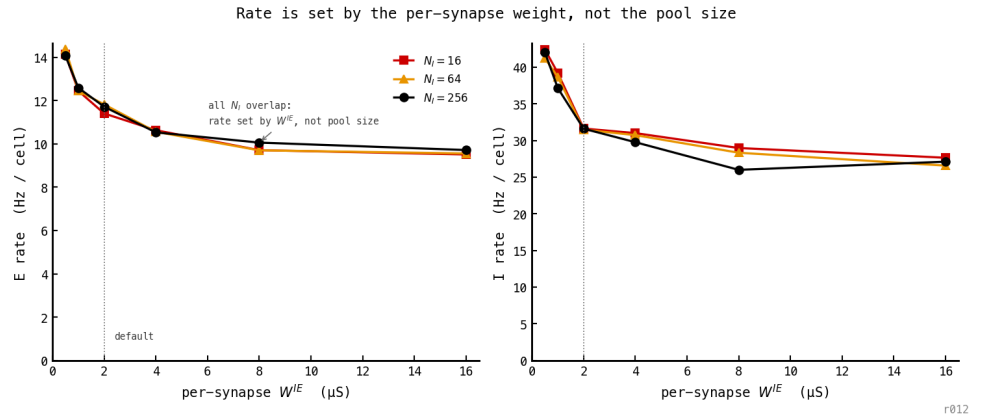


regen-per-trial-median

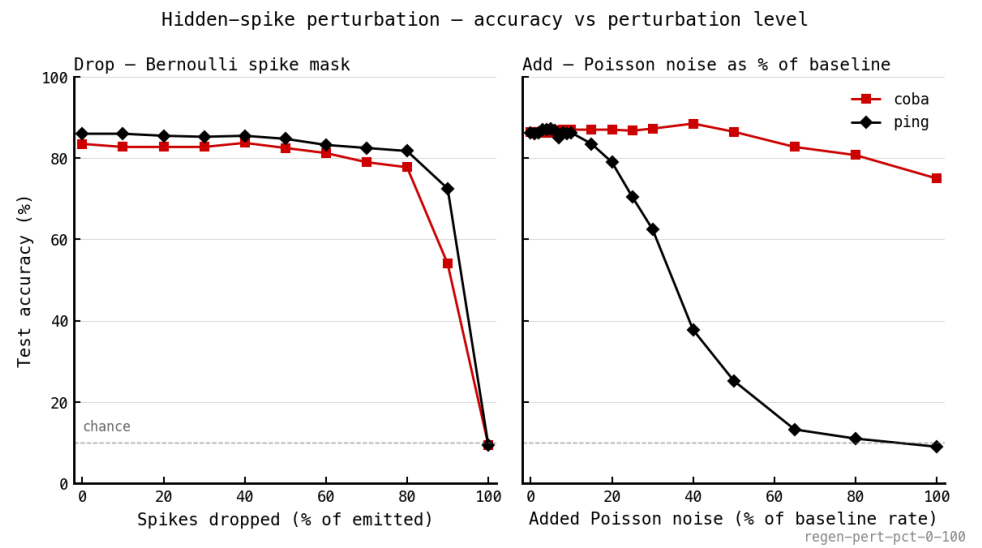
Appendix 0 Figure 6



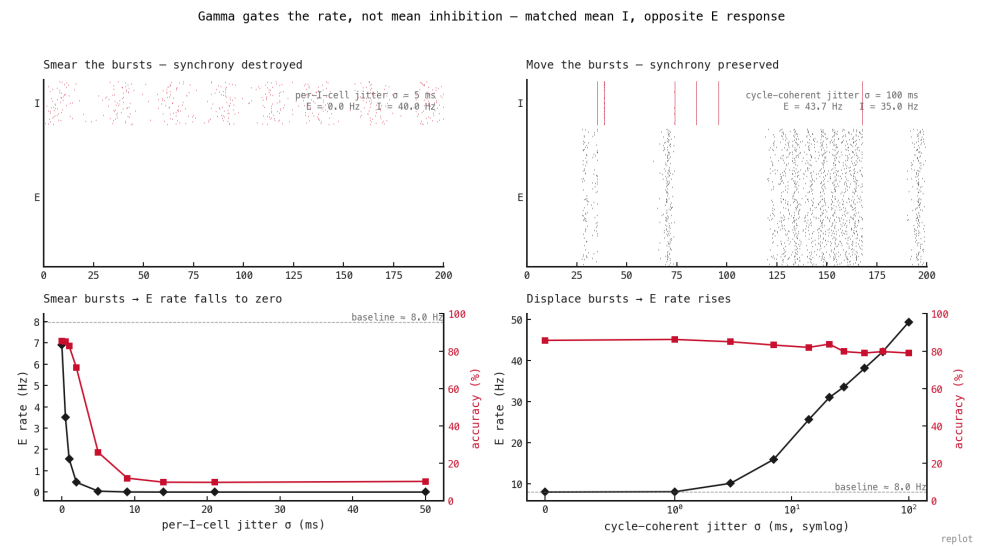
Appendix 0 Figure 7



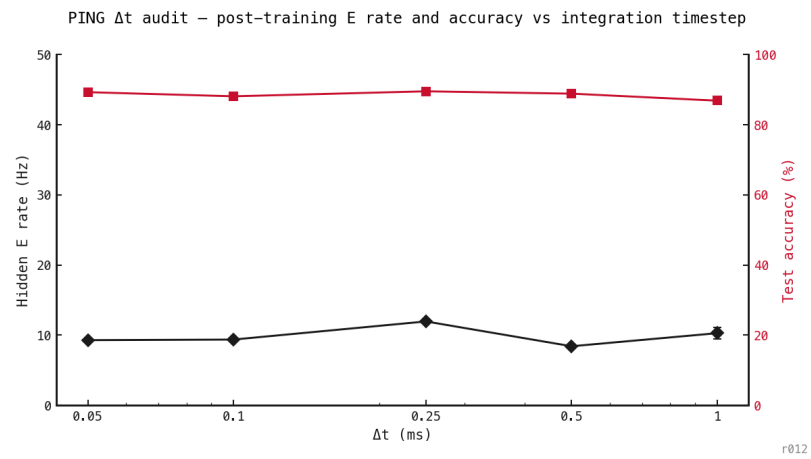
Appendix 0 Figure 8



Appendix 0 Figure 9

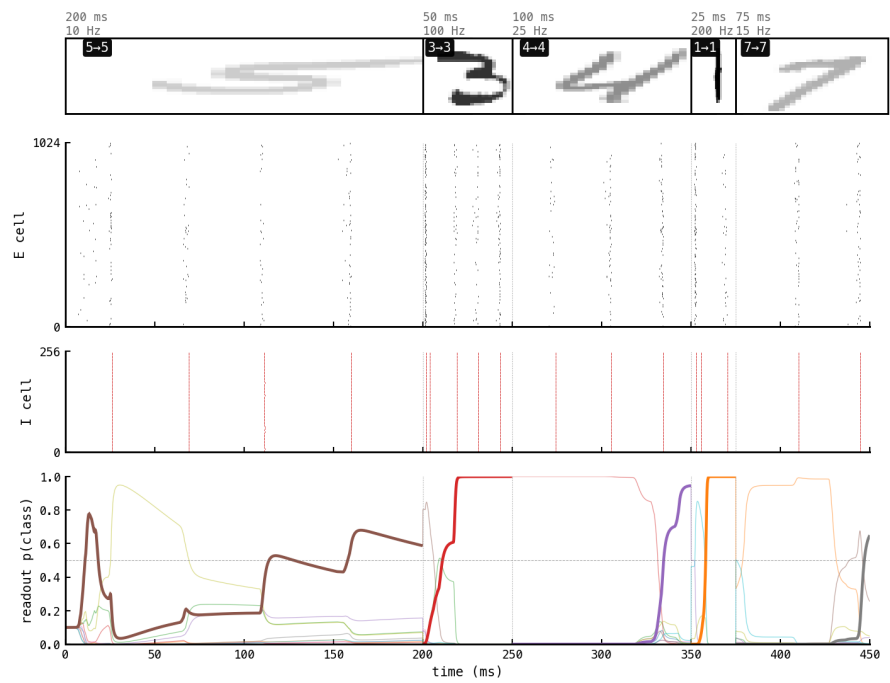


Appendix 0 Figure 10



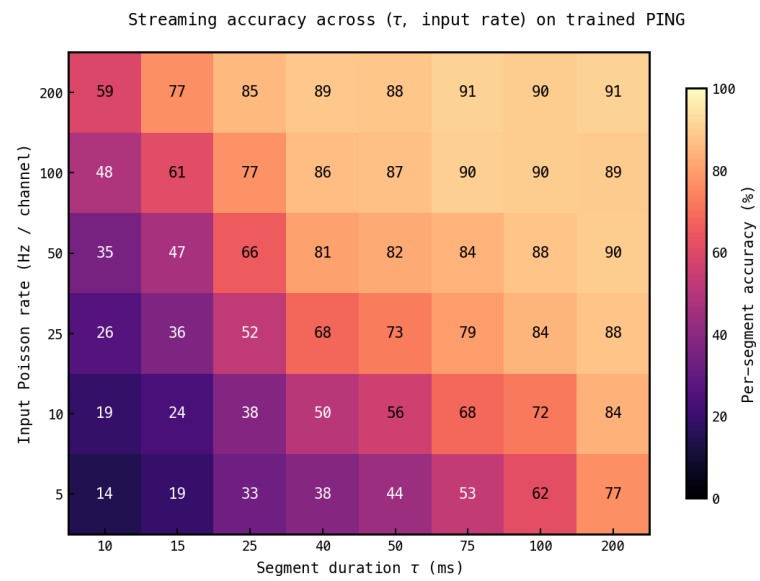
Appendix 0 Figure 11

Streaming digit classification – per-segment (τ , rate) varies, trained PING (no retraining)



nb048-replot

Appendix 0 Figure 12



r003

Appendix 0 Figure 13